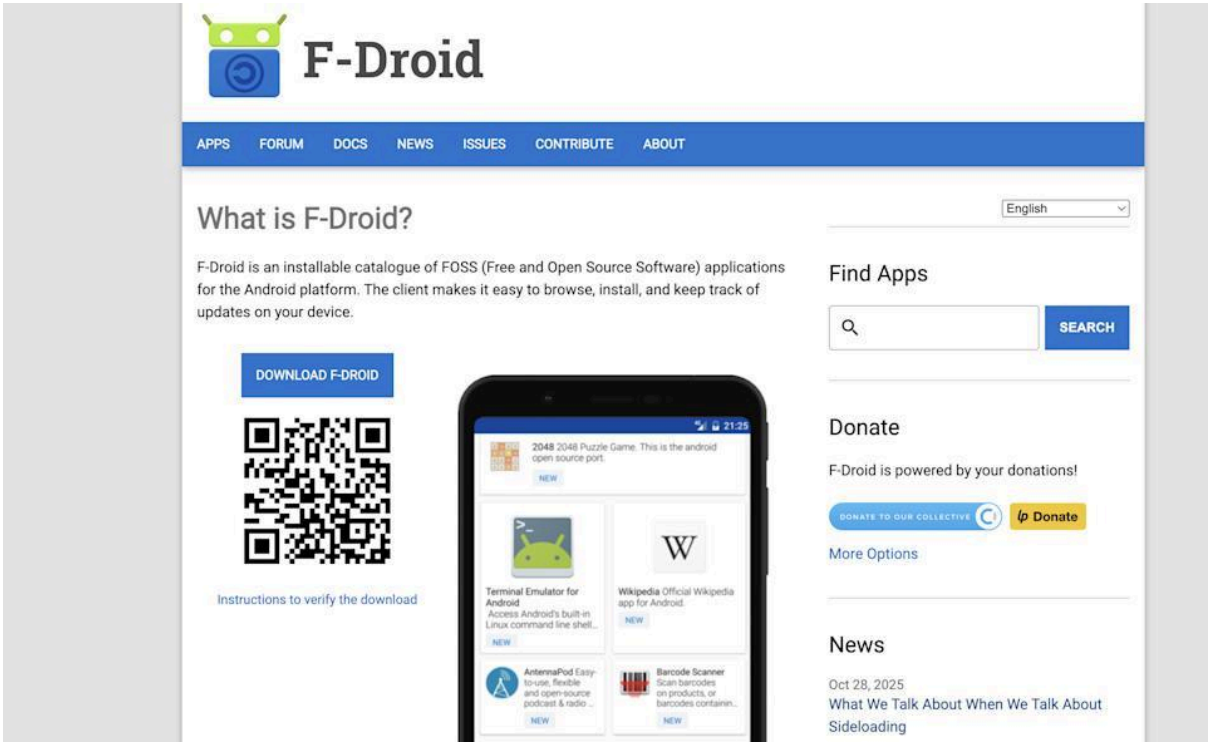
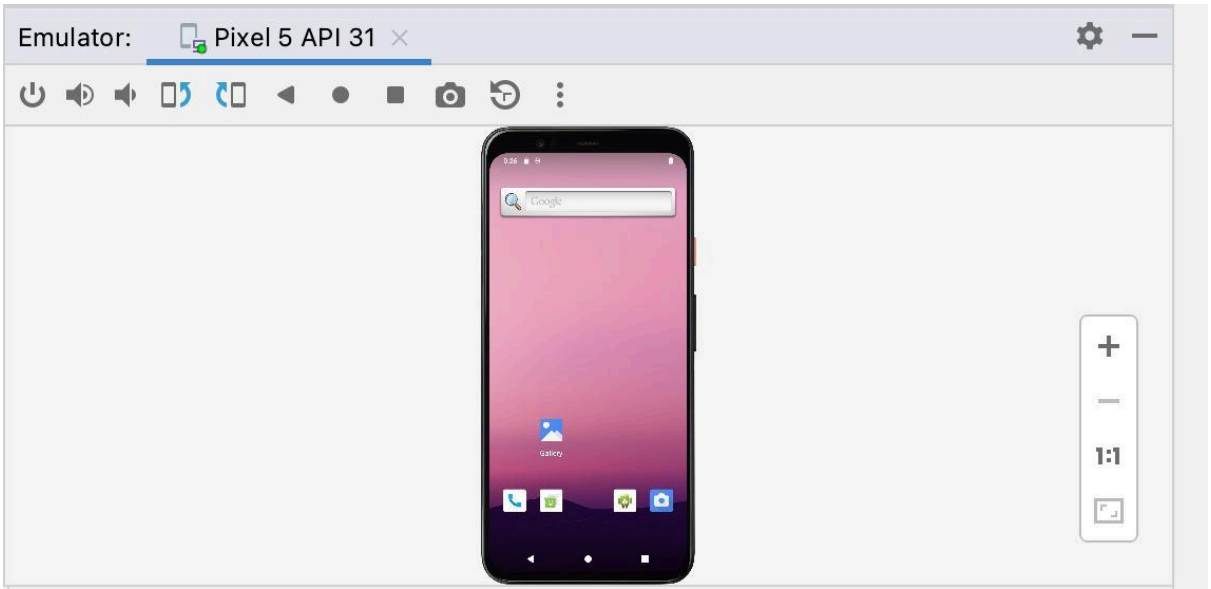
Session 3

1.Download any free APK file (for example, a simple calculator app) and install it on your Android emulator using Android Studio's Device Manager.





2.Open the installed app on your emulator, rotate the device from portrait to landscape and back, and note any UI issues or layout problems that appear.

Hint: Use the emulator's rotate button to change orientation and take screenshots if you spot any glitches.

Test orientation changes

Objective: Identify UI and layout problems when rotating the device.

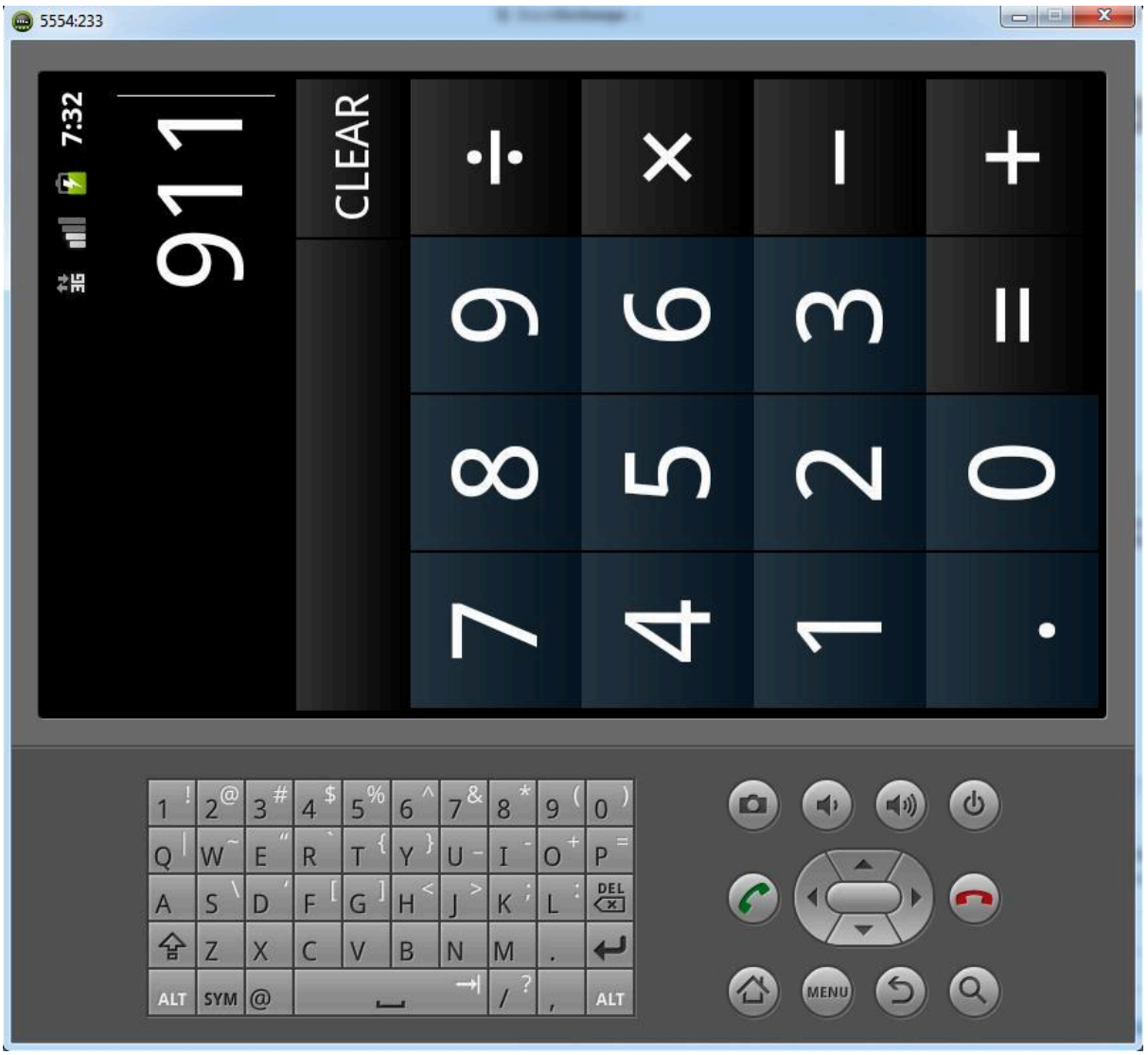
Procedure:

1. **Open the installed application in portrait mode.**
2. **Observe the screen layout, buttons, text, and menus.**
3. **Use the emulator's rotate controls to switch to landscape mode.**
4. **Check whether the UI adjusts correctly.**
5. **Rotate back to portrait mode and compare the layouts.**



Portrait mode

- **Check that buttons and text are visible.**
- **Verify that controls do not overlap.**
- **Ensure the screen is not cut off.**



Landscape mode

- **Check whether the layout adapts to the wider screen.**
- **Verify that buttons remain accessible.**
- **Check for stretched elements or excessive empty space.**

Sample observation record

Test case	Observation

Portrait to landscape	**Screen changes orientation successfully.**
Button alignment	**Some buttons may appear stretched if the layout is not responsive.**
Text visibility	**Text should remain readable and not be clipped.**
Landscape to portrait	**The layout should return to its original arrangement.**

Note: These are sample observations. Record the actual issues you see in your emulator.

3.Simulate turning off the internet connection in your emulator, then try to use a feature in the app that requires network (like fetching data or logging in). Record the error message or behavior you observe.

Test application behavior without internet

Objective: Verify how the app handles network connectivity failures.

Procedure:

1. **Open the installed app and identify a feature that requires the internet, such as login or data fetching.**
2. **Turn off Wi-Fi and mobile data in the emulator's Android settings, or use the emulator's network controls.**
3. **Try to use the network-dependent feature.**
4. **Record the error message or behavior.**
5. **Restore connectivity and retry the operation.**

You can also disable the emulator's network through ADB:

adb shell svc wifi disable

adb shell svc data disable

To re-enable Wi-Fi:

adb shell svc wifi enable

Sample observations

Test case	Expected or possible behavior
Open an online feature	Loading indicator or network error appears.
Attempt login	Login fails with a connection error.
Fetch remote data	Cached data may appear, or a retry message is shown.
Restore internet	The feature works again after retrying.

Example error message: “Network error. Please check your internet connection.”

The exact message depends on the app. Offline-capable apps may continue working using cached data.

4.Test the login or registration flow of any installed app on both your emulator and a real Android phone (if available). Compare the experience and list at least two differences you notice in speed, UI, or notifications.

Compare login or registration on an emulator and a real phone

Objective: Identify differences between virtual and physical Android devices.

Procedure:

1. **Install the same application on the emulator and a real Android phone, if available.**
2. **Open the login or registration screen on both devices.**
3. **Enter test credentials or use a test account.**
4. **Compare loading speed, keyboard behavior, UI layout, and notifications.**
5. **Record any differences.**

Comparison factor	Android emulator	Real Android phone
Speed	**Depends on computer performance and emulator configuration.**	**Depends on the phone's hardware and current system load.**

UI layout	**May differ because of virtual screen size and density.**	**Reflects the physical screen and device configuration.**
Keyboard	**Uses a virtual keyboard or computer keyboard input.**	**Uses the phone's on-screen keyboard and touch input.**
OTP verification	**SMS delivery may not work like a normal phone, depending on emulator configuration.**	**Can receive SMS if the phone, SIM, and service support it.**
Notifications	**May behave differently depending on emulator and Google Play services configuration.**	**Uses the device's actual notification and background execution environment.**

Two sample differences to report:

1. **The login screen may respond faster or slower depending on the computer and phone hardware.**
2. **OTP verification may be easier on a real phone because it can receive actual SMS messages, whereas emulator SMS support is limited or requires simulation.**

These are potential differences, not guaranteed results. Report what you observe during your test.

5.Pick any app you use daily (Instagram, Zomato, Paytm, etc.) and describe how you would test push notifications using both an emulator and a real device. Mention at least one limitation of emulators for this test.

Test push notifications on an emulator and a real device

Example app: Instagram.

Objective: Verify that notifications are delivered, displayed, and opened correctly.

Procedure:

1. **Install the app on an emulator with Google Play services support and on a real Android phone.**
2. **Log in using a test account and enable notifications in the app and Android settings.**
3. **Trigger a test notification, such as a message or another supported notification event, from a separate test account.**
4. **Check whether the notification appears when the app is in the foreground, in the background, and closed.**
5. **Tap the notification and verify that it opens the correct screen or conversation.**
6. **Repeat with the device locked, with the internet disconnected, and after connectivity is restored.**
7. **Compare notification delivery, display, sound, timing, and tap behavior on both devices.**

Sample test cases

Test scenario	Expected result
App in foreground	The app displays or handles the event according to its notification design.
App in background	A notification appears if the event and notification settings allow it.
Device locked	The notification appears according to lock-screen privacy settings.
Notifications disabled	No ordinary visible notification should appear.
Internet disconnected	Remote push delivery may be delayed until connectivity returns.
Tap notification	The app opens the relevant screen when supported.

Limitation of emulators

An emulator may not fully reproduce real-device behavior, especially for:

- **Push delivery when Google Play services are missing or incorrectly configured.**
- **Manufacturer-specific battery optimization and background restrictions.**
- **Real lock-screen behavior, Do Not Disturb settings, and hardware-specific notification sounds.**
- **Actual cellular network conditions and device-specific power management.**

For reliable results, test on both a Google Play-enabled emulator and a physical Android phone.